

MERIDIAN PARTS / OPERATIONS STRATEGY

European service-hub expansion

Board decision brief • 27 September 2026

2025 shipped sales and returns known through 31 January 2026

Recommendation: stage NLD + ESP for a reversible pilot—Spain first, with country-specific tests in the Netherlands—and release site capex only after the 90-day gates. It has the highest minimum across the supplied low/base/high/stress cases: €43,994/year floor (€166,931 base; €107,630 stress) on €270,000 year-zero capex and 6 FTE. These are scenario calculations, not a causal hub forecast.

Board choice

With no board probabilities, this uses the feasible pair with the highest minimum across the client-defined cases. That is an explicit conservative preference, not a statistical result. CZE + ESP is the strongest alternative by base contribution: EUR31,168/year more in base, but EUR75,450/year less under the defined stress. The caps are EUR450,000 and seven FTE; both options fit. Defer carries no investment and remains available.

Alternative	Capex	FTE	Low / yr	Base / yr	High / yr	Stress / yr
CZE + ESP	€225,000	5	€64,619	€198,099	€331,580	€32,180
POL + ESP	€240,000	6	€58,562	€186,329	€314,096	€29,264
POL + CZE	€205,000	5	€56,856	€178,995	€301,134	€-78,570
NLD + ESP	€270,000	6	€43,994	€166,931	€289,869	€107,630
NLD + CZE	€235,000	5	€42,289	€159,598	€276,907	€-203
NLD + POL	€250,000	6	€36,232	€147,828	€259,423	€-3,119
FRA + ESP	€340,000	7	€13,687	€129,849	€246,012	€73,105

Hard limits are board rules. Scenarios are not probability-weighted and option economics assume additive pairs with no network synergy.

1 / 2025 operating diagnosis

The shipment ledger establishes where Meridian already sells and earns contribution. It does not show how much demand a local hub would create. Modeled hub economics have two levers—added volume at current market contribution economics and assumed per-unit fulfillment savings—less annual fixed cost.

Market	Orders	Units	Net sales	Net COGS	Fulfillment	Contribution	Margin	Returned units
Germany	209	12,153	€625,792	€349,980	€13,446	€262,366	41.9%	73
France	209	11,880	€632,231	€341,453	€14,507	€276,271	43.7%	144
Netherlands	209	12,576	€697,733	€365,496	€16,593	€315,644	45.2%	144
Poland	209	13,163	€761,890	€386,698	€18,698	€356,494	46.8%	180
Czechia	209	13,733	€815,307	€397,062	€20,910	€397,335	48.7%	288
Spain	209	13,931	€850,839	€406,230	€22,708	€421,900	49.6%	252

2025 contribution by country (EUR)



Revenue, cash and contribution

Gross sales are shipped quantity × price less discounts. Refund credits reduce net sales. Net sales approximate booked revenue under the supplied credit ledger, not cash collected: payment timing, settlement FX, fees and working capital were unavailable. Contribution further deducts net COGS after recovered restocked units and nonrefundable fulfillment. It is not EBITDA or hub profit; option recurring costs are deducted separately.

Country margins equal contribution ÷ net sales. Monthly records contain 72 rows (six countries × 12 months); each annual country total reconciles to its monthly rows.

2 / Reconciliation and source quality

The client's transaction, cost and option data are synthetic. They were collected from the frozen local source room; no live client account, customer or interview was accessed. Raw inputs and source-room metadata are retained under `sources/raw`.

Orders: 1,328 raw rows represent 1,297 IDs. 13 exact duplicate rows were removed; 18 IDs contain distinct revisions; no same-revision conflicts were found. Highest numeric revision wins across both extract pages and the whole-row correction file. 1,254 shipped, non-test, 2025 orders are included; 43 selected records are excluded (`{'not_shipped': 24, 'test_shipped': 18, 'shipped_date_outside_2025_or_missing': 1}`). Zero-price shipment orders are retained (12); January 2026 shipments are not applied to the base.

Returns: 264 rows represent 243 IDs. 20 exact duplicate rows removed; 1 ID had multiple revisions; none have conflicting same-revision rows. The highest revision is kept. Through the inclusive 31 January 2026 cutoff, 241 returns linked to eligible selected sales are included. 1 orphan/ineligible record is quarantined (R-ORPHAN); 1 record arrived after cutoff. Return consistency checks: 0 orders exceed shipped units, 0 return rows restock more units than returned, and 0 orders refund more than gross local sales.

FX and rounding

For PLN and CZK, sales and refund credits use the arithmetic mean of published daily ECB reference observations in the original shipment month, quoted as local currency units per EUR; EUR is 1. No inversion or current FX is used. Each order's gross, refund, gross COGS, recovered COGS and fulfillment amounts are rounded half-up to cents before summation. These reference quotes are analytical translation assumptions, not transaction rates.

Missing costs: 0; missing FX: 0; missing World Bank population cells: 0; missing GDP cells: 0. No values are imputed.

3 / Market context

Archived official World Bank series provide population (persons) and GDP per capita (current US dollars) for 2022–2024. Population change and income level contextualize scale and economic environment only; neither is direct evidence of Meridian product demand.

Country	Population 2022	Population 2023	Population 2024	Change 2022–24	GDP/capita 2024, current USD
Germany	83,177,813	83,287,273	83,516,593	0.4%	\$56,104
France	68,184,457	68,372,286	68,551,653	0.5%	\$46,103
Netherlands	17,700,982	17,877,117	17,993,485	1.7%	\$67,465
Poland	36,821,749	36,687,353	36,559,233	-0.7%	\$25,104
Czechia	10,672,118	10,864,042	10,905,028	2.2%	\$31,828
Spain	47,786,102	48,352,528	48,848,840	2.2%	\$35,327

All 18 archived year-country cells are present. Population values are persons. GDP per capita is nominal current USD—not PPP and not constant-price real income. Historical public series were collected into a frozen source room on 27 September 2026 and may incorporate revisions after the measured 2025 trading year. Original API URLs, response hashes and vintage are listed in `sources/source-register.json`.

Market scope limitation

No market sizing, competitor comparison, localized demand survey, labor cost benchmark, real-estate quote, service-level history or regulatory diligence was supplied or added. This recommendation does not infer product fit from GDP per capita or population.

4 / Hub economics and decision

For each country, annual incremental contribution equals $C \times u + U \times (1+u) \times s - F$: C is 2025 contribution, U shipped units, u is assumed volume uplift (10%, 25%, 40%), s is client-assumed fulfillment saving per unit and F recurring annual fixed cost. Capex K is separate year-zero investment; simple undiscounted payback is $K \div$ positive annual increment, otherwise null.

Stress is the defined joint sensitivity, not a forecast probability: base uplift 25%, refund shock 3% of gross sales without added cost recovery, plus 10% of net sales depreciation for PLN/CZK options. Stressed contribution is compared against the unchanged normal baseline, then uplifted savings and recurring cost are applied. Pairs add country economics without synergy.

Option	Capex yr 0	Fixed / yr	FTE	Low / yr	Base / yr	High / yr	Stress / yr	Base payback
NLD + ESP	€270,000	€120,000	6	€43,994	€166,931	€289,869	€107,630	1.6 years
CZE + ESP	€225,000	€95,000	5	€64,619	€198,099	€331,580	€32,180	1.1 years
POL + ESP	€240,000	€100,000	6	€58,562	€186,329	€314,096	€29,264	1.3 years
POL + CZE	€205,000	€85,000	5	€56,856	€178,995	€301,134	€-78,570	1.1 years
NLD + CZE	€235,000	€105,000	5	€42,289	€159,598	€276,907	€-203	1.5 years

The strongest different feasible alternative is CZE + ESP: €198,099/year base against €166,931/year for NLD + ESP—a modeled difference of €-31,168/year. The stress values are €32,180 and €107,630/year respectively. It uses €225,000 capex and 5 FTE, compared with €270,000 and 6 FTE. Defer avoids all capex and recurring cost while foregoing modeled increment.

Switching condition

The choice assumes 10%–40% volume uplift and the stated per-unit savings. For one option, the break-even uplift is $(F - U \times s) / (C + U \times s)$, when the denominator is positive. If a reversible pilot shows savings are not avoidable, uplift misses break-even, or annual fixed cost rises enough to erase contribution, defer, reconfigure, or select the runner-up. Validate all levers before a full lease or site commitment.

Preference provenance: selecting the combination with the highest minimum across low/base/high/stress is a conservative judgment because no board probabilities or weights were supplied. A base-only choice selects CZE + ESP.

5 / First 90 days

Stage investment so the board can stop after learning whether customer and cost response support the modeled economics.

Timing / owner	Work, dependencies and decision gate
Days 0–15 COO + Finance	Reconcile 2025 cohort and return baseline; validate invoice-level avoidable fulfillment cost, option cost assumptions, pilot SKU/zone, comparison design and KPI definitions. Gate 1: auditable baseline and target cohort before any site commitment.
Days 16–35 Operations + Data	Cost and design the service zone, inventory/transfer process, staffing, temporary 3PL/space, country legal/tax review and service SLA. Gate 2: signed costed design, data coverage $\geq 98\%$, capex and FTE inside board caps.
Days 36–65 Country lead + Supply Chain	Run a reversible, limited-SKU pilot with temporary space/3PL or reserved stock; monitor matched/holdout cohorts, delivery promise, fill rate, contribution and handling cost. Gate 3: require at least four weeks usable data; stop for legal/safety breach, stock accuracy $< 98\%$, or negative contribution trend.
Days 66–90 CFO + COO + Board sponsor	Compare to pre-registered baseline; update scenarios, quotes, FX/returns stress, recurring costs and staffing. Gate 4: release site capex only if base annual contribution remains positive after fixed cost, all caps hold, and downside is accepted; otherwise defer/reconfigure.

KPI plan: weekly pilot and monthly board reporting for incremental shipped orders/units, net sales and return rate by sale cohort, contribution after returns, avoidable fulfillment EUR/unit, median/P90 delivery time, on-time-in-full, fill rate, stock accuracy, restocked share and return cycle time, capex committed and staffed FTE. Publish numerator, denominator, comparison period and data coverage. Do not call seasonal/mix changes hub lift without a credible comparison.

Principal risks: demand does not rise; savings are not truly avoidable; reference FX differs from settlement; inventory fragmentation increases stockouts or working capital; returns erode margins; local fixed costs, labor, lease or compliance assumptions are incomplete. Mitigations are a matched reversible pilot, invoice checks, monthly settlement-FX monitoring, narrow SKU scope and local quotes/legal review before commitment.

Appendix / ECB reference-rate means

Arithmetic means of published business-day reference rates during each 2025 calendar month. Quote: local currency units per EUR. Daily observation counts vary with the calendar and ECB publication availability. EUR-denominated markets use 1.00.

Month	PLN per EUR	Days	CZK per EUR	Days
2025-01	4.246655	22	25.163318	22
2025-02	4.172225	20	25.076650	20
2025-03	4.182010	21	25.001238	21
2025-04	4.265220	20	25.038950	20
2025-05	4.253800	21	24.922857	21
2025-06	4.265771	21	24.804286	21
2025-07	4.254148	23	24.624609	23
2025-08	4.261305	21	24.516619	21
2025-09	4.258886	22	24.346773	22
2025-10	4.248791	23	24.315391	23
2025-11	4.237635	20	24.234050	20
2025-12	4.223857	21	24.259286	21

Source: archived ECB daily reference rates from ecb-history.zip and its lossless ecb-history.csv extraction in sources/raw. The source register records original URL, retrieval timestamp, hashes, bytes, units and coverage. Public series is an official snapshot; the synthetic client transaction data are distinct from these observations.

Reproducibility: run **python3 analysis/build.py**, then **python3 analysis/render_pdfs.py** from the project root. The metrics JSON is the canonical exchange dataset; the workbook includes all monthly and country calculations, scenarios, context, sources and quality notes.

Limits: one synthetic trading year; no demand causal design, discounted cash flow, ramp-up, synergies, tax, lease exit, working capital or probability model. Public historical revisions may post-date the trading year. Recommendation is conditional; it is not statistically proven.